Validating Data for a Linguistic Resource

Error analysis, agreement, and giving back

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JPC3 — Improving the Czech Wordnet

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Overview

- Why be sure about the data
- Error analysis: beyond one number
- Why two annotators
- Cohen's κ, worked through
- Giving fixes back to ARASAAC, Wikidata, …

Roadmap

1. Why be sure?
2. The new bottleneck
3. Error analysis
4. Agreement
5. Cohen's kappa
6. Giving back

Why be sure?

- A wordnet is a **trusted source**
- Many tools read it; few check it
- An error you add, everyone inherits
- Once released, it is copied — hard to recall
- The Czech wordnet feeds OMW, sense tagging, AAC, …

The cost of one wrong link

- Project 4 links pictographs for **AAC** users
- A wrong synset tag $\rightarrow$ the wrong picture
- A child who cannot speak is handed the wrong word
- Here a lexical slip is not cosmetic — it misleads

Precision is not pedantry when someone relies on the data

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Understanding Is the Bottleneck

- The starter code generates candidates in seconds
- Machines already verify themselves well
- The slow part is **understanding** what came out
- “Cognitive debt”: skip it and you lose the plot
- Generation got cheap; understanding did not (Litt, 2026)

https://www.geoffreylitt.com/2026/07/02/understanding-is-the-new-bottleneck.html

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Staying in the Loop

Three ways to keep understanding ahead of the machine:

- Explain before you trust: write what the data *says*
- Visualise it — a figure you can poke, not just read
- Quiz each other on a sample — a **speed regulator**

Augmentation, not automation — get deeper in the loop

Look at Your Data as You Go

- The final review is not the only check
- Look at the output at every step, not just the end
- Dump it in a **viewable format** — TOML, CSV, a plot
- A wrong turn caught early is cheap to undo
- A literate program shows its output inline, by habit

“Actually look at your data!”

Explainers as Literate Programming

- Knuth (1984): write code for **humans** to read
- Prose and code woven together, explanation first
- Litt's explainer is the same idea — authorship flipped
- The agent writes the code; you write the understanding
- Your report is the literate layer over the scripts

Revived for LLMs: "Natural Language Outlines" (2024), "Verifiable Literate Programming" for validating generated code (2026)

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Error Analysis

- Take a **random** sample (aim for 200)
- Judge each by hand: right or wrong?
- Precision = correct / judged
- But the single number is the least of it
- Group the mistakes into **types** — that is the payoff
- One real example per type, with a count

Running example: přechylování

Candidate: add a feminine form to a role-noun synset.

(1) Czech [ces]
učitel → *učitel-ka*
teacher.M teacher-F
“(male) teacher → (female) teacher”

- 1,919 role-noun synsets to work from (`noun.person`)
- Easy to generate; the question is which are **right**

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An Error Taxonomy

A sample of 200 generated feminine candidates:

Error type	Count	Example
Correct	171	*lékař* → *lékařka*
Coincidental suffix	12	*past* (not a base + *-ka*)
Wrong sense	9	feminine of a non-personal reading
Already present	8	lemma was in the synset

Precision 0.86 — but the four rows tell the next team where to look

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Why Two Annotators?

- One judge can be **consistently** wrong
- A second judge tests whether the scheme is learnable
- Agreement $\approx$ reliability of the guidelines
- You work in pairs — so measure it
- Report how you resolved the disagreements

Raw Agreement Is Not Enough

- “We agreed 85% of the time” sounds good
- But some agreement happens by **chance**
- Two coin-flippers agree half the time
- If almost everything is “correct”, agreeing is easy
- Correct for chance $\rightarrow$ Cohen’s κ

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Cohen's Kappa

Agreement above what chance would give:

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

- p_o: observed agreement (the proportion you both called the same)
- p_e: expected agreement **by chance**, from each annotator's rates
- $\kappa = 1$ perfect; $\kappa = 0$ no better than chance

A Worked Example

200 candidates, two annotators, good vs. bad:

	B: good	B: bad	total
A: good	130	20	150
A: bad	10	40	50
total	140	60	200

- $p_o = (130 + 40)/200 = 0.85$
- $p_e = 0.75 \times 0.70 + 0.25 \times 0.30 = 0.60$

$$\kappa = \frac{0.85 - 0.60}{1 - 0.60} = \mathbf{0.63}$$

Reading Kappa

- Rough scale (Landis & Koch, 1977): > 0.8 very good, 0.6–0.8 substantial
- Our 0.63 is **substantial** — the guidelines mostly work
- Low κ? Fix the guidelines, then re-annotate
- Beware: few categories or a skewed split inflate it
- Report κ *and* the raw table, never κ alone

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Why Open? — Not Just the Warm Inner Glow

- Our goal: a better Czech wordnet — *and* a **contribution**
- Open data others can build on, not a private copy
- The warm inner glow is nice, but not the point
- The point: your fixes reach every downstream user
- A result that outlives the course

Feedback Upstream

Every error you find is a **gift** to the source.

- ARASAAC: report wrong synset tags; send Czech keywords back
- Wikidata: add or fix the ILI (P5063) and WordNet id (P8814)
- Global Symbols / ARAWN: contribute the matches you validated
- CILI: report synsets with no ILI; DeriNet: report gaps
- PDT: the unmapped residue is data for the maintainers

Give Back: ARASAAC

Project 4 — pictographs for AAC.

- Found a **wrong** synset tag? Report it to ARASAAC
- Missing Czech keyword? Submit the translation
- Note colour and gender variants you matched
- Their pictograms carry synsets — a fix helps every language

Give Back: Wikidata

Project 5 — Wikidata → ILI → Czech.

- Only a WordNet id (P8814)? Add the **ILI** (P5063)
- Linked to the wrong sense? Correct the statement
- Confirmed a Czech label? Add it, with a reference
- Cite the wordnet as the source — statements need one

Give Back: PDT and CILI

Project 3 — the sense-tag residue.

- The **unmapped** tags are a bug list for the maintainers
- Split or merged senses? Propose the current synset
- Share your 1.5 → current mapping table
- New concept with no ILI? Ask CILI to mint one

Give Back: DeriNet

Project 2 — derivational relations.

- Validated feminine-motion pairs? Offer them back
- A family the wordnet **lacks**? Flag the gap
- A DeriNet error you caught? Report it
- Both resources improve from the same checking

The Virtuous Cycle

- Open data gets better only when fixes **flow back**
- A pull request or an issue beats a private fix
- Cite what you used; credit who made it
- Release under an open licence, loadable with `wn`
- Next year's class starts from your cleaner data

Every new wordnet makes the network richer

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Conclusion

- Extraction is scaffolded — **validation** is the work
- Sample, judge, and report precision
- Categorise the errors; the taxonomy guides the next step
- Two annotators, and κ with the table
- Push every fix back upstream

Thanks

- Validate carefully — others will trust your data
- Actually look at your data!
- Code and template: the JPC3 repository

Questions welcome — and so are pull requests

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